

ASEN 3801 Lab 5: Fixed-Wing Aircraft Simulation and Control

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INTRODUCTION

This lab is designed to calculate the aerodynamic forces and moment from the state vectors, control surfaces, and aerodynamic coefficients. The application of MATLAB would be used to simulate the flight of fixed wing aircraft. The use of MATLAB to get simulations for time 3 seconds and 200 seconds separately for Fixed-Wing aircraft plotted for all 12 state vectors including control inputs and trajectory path. Lastly, the third part of the lab describes the aircraft behavior for longitudinal response for short period (time span 3 seconds) and phugoid mode (100 seconds).

PROBLEM 1: AIRCRAFT DYNAMICS PLOTTING FUNCTION

1. Updating Plotting Function

After updating our plotting function to include aileron, elevator, and rudder deflection as control input variables, we have the following:

```
function PlotAircraftSim(time, aircraft_state_array, control_input_array, fig, col)
% PlotAircraftSim - Lab 5
t = time(:);
n = numel(t);
if size(aircraft_state_array,2) ~= n
    aircraft_state_array = aircraft_state_array.';
end
if size(control_input_array,2) ~= n
    control_input_array = control_input_array.';
end
de_deg = rad2deg(control_input_array(1,:));
da_deg = rad2deg(control_input_array(2,:));
dr_deg = rad2deg(control_input_array(3,:));
dt = control_input_array(4,:);
%% Figure 1 - Inertial Position
figure(fig(1));
subplot(3,1,1); hold on; plot(t, aircraft_state_array(1,:), col); ylabel('x (m)'); grid on;
subplot(3,1,2); hold on; plot(t, aircraft_state_array(2,:), col); ylabel('y (m)'); grid on;
subplot(3,1,3); hold on; plot(t, -aircraft_state_array(3,:), col); ylabel('z (m)'); xlabel('Time (s)'); grid
on;
sgtitle('Inertial Position (Fixed-Wing)');
%% Figure 2 - Euler Angles
figure(fig(2));
subplot(3,1,1); hold on; plot(t, rad2deg(aircraft_state_array(4,:)), col); ylabel('\phi (deg)'); grid on;
subplot(3,1,2); hold on; plot(t, rad2deg(aircraft_state_array(5,:)), col); ylabel('\theta (deg)'); grid on;
subplot(3,1,3); hold on; plot(t, rad2deg(aircraft_state_array(6,:)), col); ylabel('\psi (deg)'); xlabel('Time
(s)'); grid on;
sgtitle('Euler Angles (Fixed-Wing)');
%% Figure 3 - Body-Frame Velocity
figure(fig(3));
subplot(3,1,1); hold on; plot(t, aircraft_state_array(7,:), col); ylabel('u (m/s)'); grid on;
subplot(3,1,2); hold on; plot(t, aircraft_state_array(8,:), col); ylabel('v (m/s)'); grid on;
```

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subplot(3,1,3); hold on; plot(t, aircraft_state_array(9,:), col); ylabel('w (m/s)'); xlabel('Time (s)'); grid
on;
sgtitle('Body-Frame Velocity (Fixed-Wing)');
%% Figure 4 - Angular Velocity
figure(fig(4));
subplot(3,1,1); hold on; plot(t, rad2deg(aircraft_state_array(10,:)), col); ylabel('p (deg/s)'); grid on;
subplot(3,1,2); hold on; plot(t, rad2deg(aircraft_state_array(11,:)), col); ylabel('q (deg/s)'); grid on;
subplot(3,1,3); hold on; plot(t, rad2deg(aircraft_state_array(12,:)), col); ylabel('r (deg/s)'); xlabel('Time
(s)'); grid on;
sgtitle('Angular Velocity (Fixed-Wing)');
%% Figure 5 - Control Inputs (per Lab 5)
figure(fig(5));
subplot(4,1,1); hold on; plot(t, de_deg, col); ylabel('\delta_e (deg)'); grid on;
subplot(4,1,2); hold on; plot(t, da_deg, col); ylabel('\delta_a (deg)'); grid on;
subplot(4,1,3); hold on; plot(t, dr_deg, col); ylabel('\delta_r (deg)'); grid on;
subplot(4,1,4); hold on; plot(t, dt, col); ylabel('\delta_t (0-1)'); xlabel('Time (s)'); grid on;
sgtitle('Control Inputs: Surfaces in Degrees, Throttle as Fraction');
%% Figure 6 - 3D Path
figure(fig(6));
hold on;
plot3(aircraft_state_array(1,:), aircraft_state_array(2,:), -aircraft_state_array(3,:), col, 'LineWidth', 1.2);
plot3(aircraft_state_array(1,1), aircraft_state_array(2,1), -aircraft_state_array(3,1), 'go',
'MarkerFaceColor', 'g');
plot3(aircraft_state_array(1,end), aircraft_state_array(2,end), -aircraft_state_array(3,end), 'ro',
'MarkerFaceColor', 'r');
grid on; axis vis3d;
xlabel('x (m)'); ylabel('y (m)'); zlabel('Altitude (m)');
view(3);
title('3D Aircraft Path (Fixed-Wing)');
end

```

PROBLEM 2: AIRCRAFT EOM FUNCTION

```

function xdot = AircraftEOM(time, aircraft_state, aircraft_surfaces, wind_inertial, aircraft_parameters)
ap = aircraft_parameters;
xi = aircraft_state(1);
yi = aircraft_state(2);
zi = aircraft_state(3);
phi = aircraft_state(4);
theta = aircraft_state(5);
psi = aircraft_state(6);
u = aircraft_state(7);
v = aircraft_state(8);
w = aircraft_state(9);
p = aircraft_state(10);
q = aircraft_state(11);
r = aircraft_state(12);
[~, ~, ~, rho] = atmosisa(-zi); %Negative to account for us putting negative values into initial call
[aero_forces, aero_moments] = AeroForcesAndMoments(aircraft_state, ...
                                                    aircraft_surfaces, ...
                                                    wind_inertial, ...
                                                    rho, ...
                                                    aircraft_parameters);

X = aero_forces(1);
Y = aero_forces(2);
Z = aero_forces(3);
L = aero_moments(1);
M = aero_moments(2);
N = aero_moments(3);
m = ap.m;
g = ap.g;
Ix = ap.Ix;
Iy = ap.Iy;
Iz = ap.Iz;
Ixz = ap.Ixz;
udot = r*v - q*w + X/m - g*sin(theta);
vdot = p*w - r*u + Y/m + g*sin(phi)*cos(theta);

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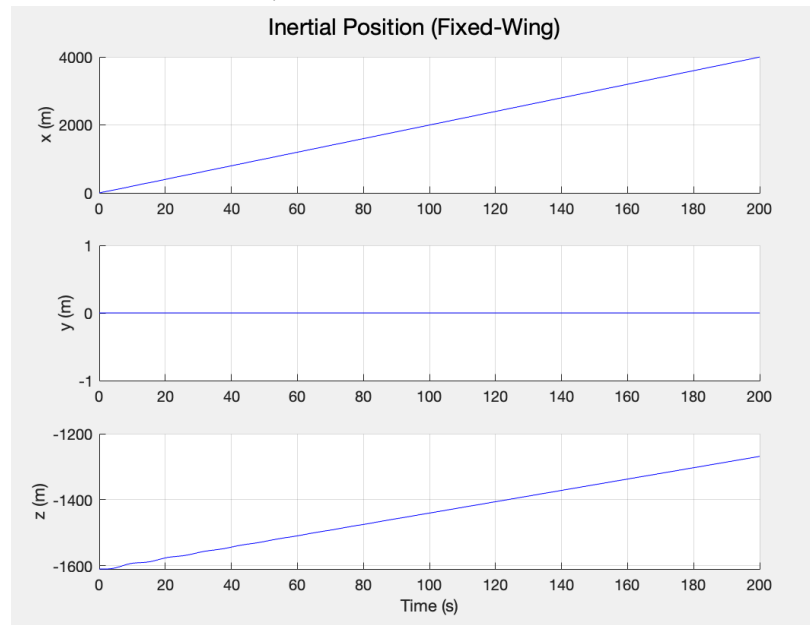
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wdot = q*u - p*v + Z/m + g*cos(phi)*cos(theta);
Gamma = Ix*Iz - Ixz^2;
Gamma1 = (Ixz*(Ix - Iy + Iz))/Gamma;
Gamma2 = (Iz*(Iz - Iy) + Ixz^2)/Gamma;
Gamma3 = Iz/Gamma;
Gamma4 = Ixz/Gamma;
Gamma5 = (Iz - Ix)/Iy;
Gamma6 = Ixz/Iy;
Gamma7 = (Ix*(Ix - Iy) + Ixz^2)/Gamma;
Gamma8 = Ix/Gamma;
pdot = Gamma1*p*q - Gamma2*q*r + Gamma3*L + Gamma4*N;
qdot = Gamma5*p*r - Gamma6*(p^2 - r^2) + M/Iy;
rdot = Gamma7*p*q - Gamma1*q*r + Gamma4*L + Gamma8*N;
phi_dot = p + q*sin(phi)*tan(theta) + r*cos(phi)*tan(theta);
theta_dot = q*cos(phi) - r*sin(phi);
psi_dot = q*sin(phi)*sec(theta) + r*cos(phi)*sec(theta);
cphi = cos(phi); sphi = sin(phi);
cth = cos(theta); sth = sin(theta);
cpsi = cos(psi); spsi = sin(psi);
R_bi = [ cth*cpsi, sphi*sth*cpsi - cphi*spsi, cphi*sth*cpsi + sphi*spsi;
         cth*spsi, sphi*sth*spsi + cphi*cpsi, cphi*sth*spsi - sphi*cpsi;
         -sth, sphi*cth, cphi*cth ];
vel_inertial = R_bi * [u; v; w];
xidot = vel_inertial(1);
yidot = vel_inertial(2);
zidot = vel_inertial(3);
xdot = zeros(12,1);
xdot(1) = xidot;
xdot(2) = yidot;
xdot(3) = zidot;
xdot(4) = phi_dot;
xdot(5) = theta_dot;
xdot(6) = psi_dot;
xdot(7) = udot;
xdot(8) = vdot;
xdot(9) = wdot;
xdot(10) = pdot;
xdot(11) = qdot;
xdot(12) = rdot;
end

```

1. Simulation In Initial State 1

When the simulation is run for 3 seconds, our results are:



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Fig 2.11: Fixed-wing aircraft Inertial [x,y,z] positions

Our displacement along the inertial axes is consistent with our undisturbed state. Because the aircraft has no lateral initial velocity or disturbances, the displacement along the inertial y-axis is zero for the duration of the simulation. The aircraft flies due North at 21 meters per second, which is exactly the slope of inertial x-axis displacement versus time. Our altitude also shows a slow downwards descent, this is because it is flying at an insufficient velocity and with no pitch to balance the lift equations and fully counteract gravity, therefore it maintains a smooth linear downwards glide at roughly 12 meters per second, the visual “upwards” trajectory is merely an artifact of the convention, it should be understood that the aircraft is losing altitude and from the perspective of a viewer on the ground would exhibit a downwards trajectory.

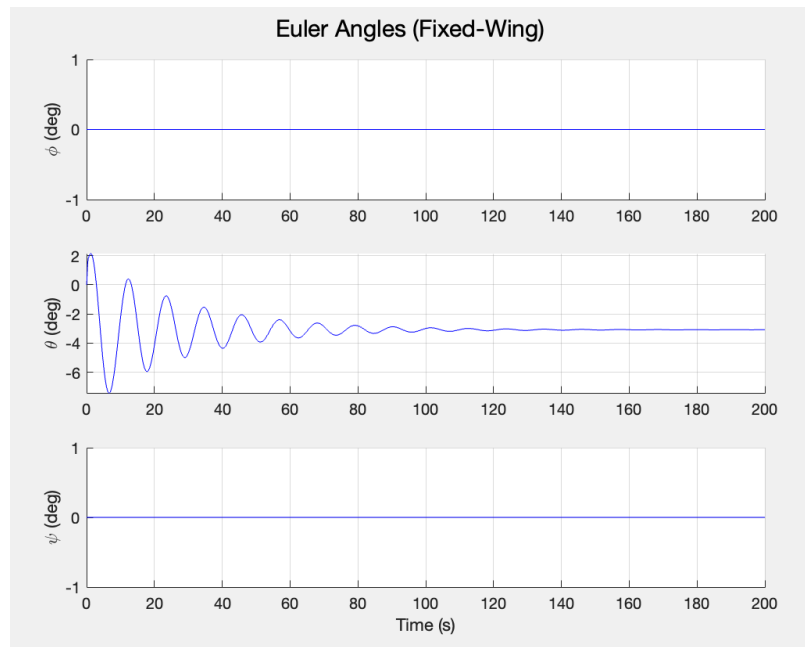


Fig 2.12: Fixed-wing aircraft Euler angles

The Euler angles of the aircraft are consistent with our given initial conditions; both roll and yaw remain undisturbed at zero degrees for the duration of the simulation. The pitch oscillates up and down, with an initial amplitude of roughly two degrees, a period of roughly 10 seconds, and dies out almost completely by 100 seconds. This pitch behavior is consistent with the phugoid mode glide downward.

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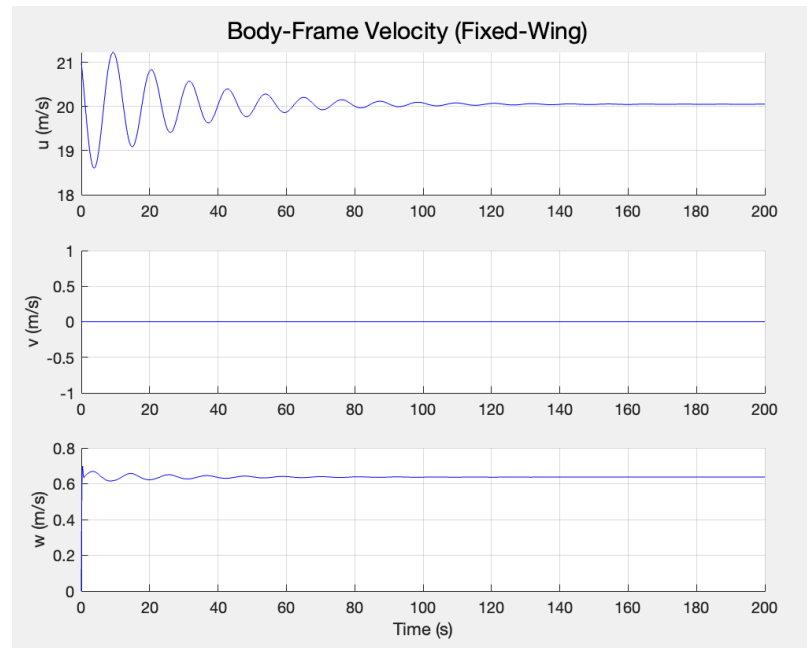


Fig 2.13: Fixed wing aircraft Body-Frame $[u,v,w]$ component velocity

The velocity $u(t)$ has a damped oscillation with an approximately 1 meters per second amplitude. It takes about 120 seconds to settle to 20 meters per second. $v(t)$ remains constant at zero for the duration of the simulation, which is consistent with no lateral components. $w(t)$ exhibits very minor oscillations which die out by 60 seconds, and is fairly constant at approximately 0.63 meters per second, consistent with a gradual decline. Note that because of the sign convention when reporting altitude, a positive $w(t)$ corresponds to losing altitude.

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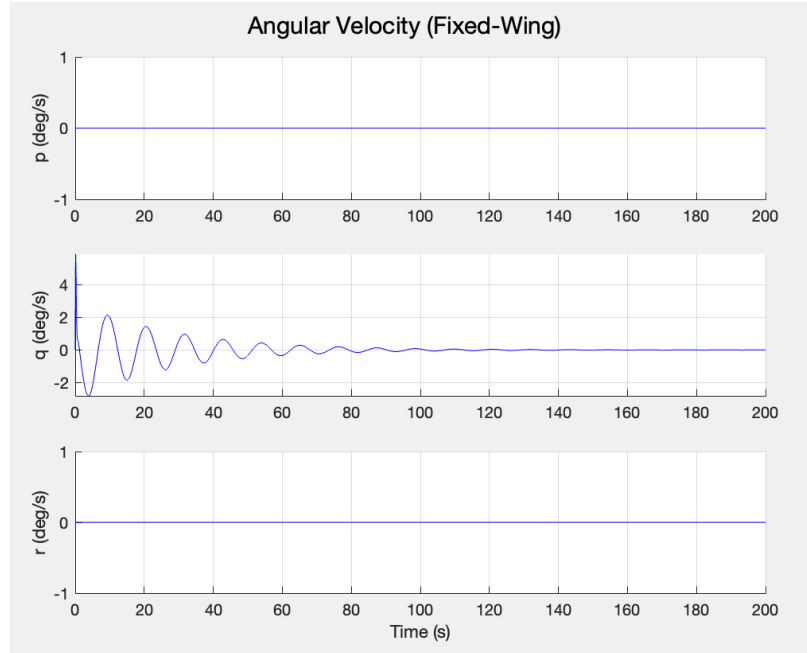


Fig 2.14: Fixed-wing Angular velocities $[p,q,r]$ components

Yaw rate and roll rate are both zero for the duration of the flight, this is consistent with steady level flight, the pitch rate oscillations are consistent with the oscillations in pitch visible in Fig 2.12 and with the overall “bobbing” motion of the plane on its glide downward.

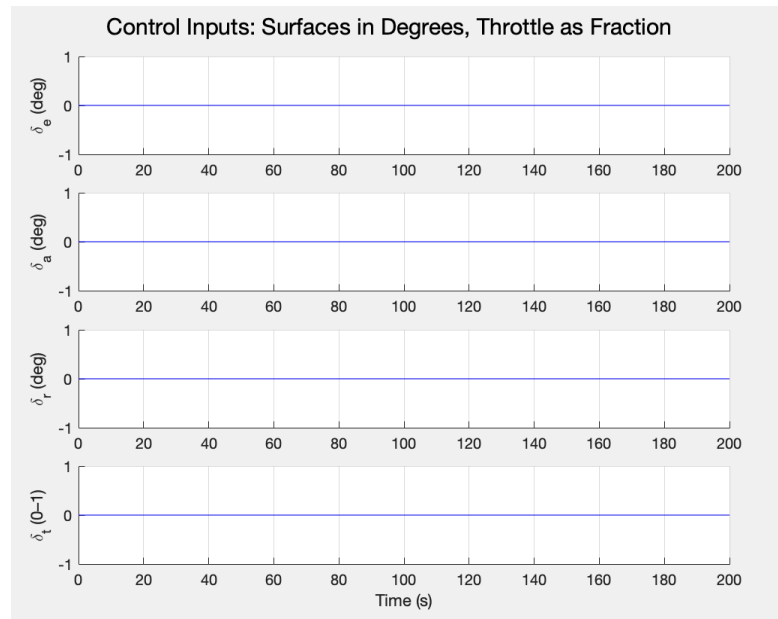


Fig 2.15: Aircraft Control Inputs

All control inputs remain at zero for the duration of the simulation, this is consistent

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with the lack of a control law and no initial control inputs. These input values are independent of the the actual performance of the aircraft unless a control law is implemented.

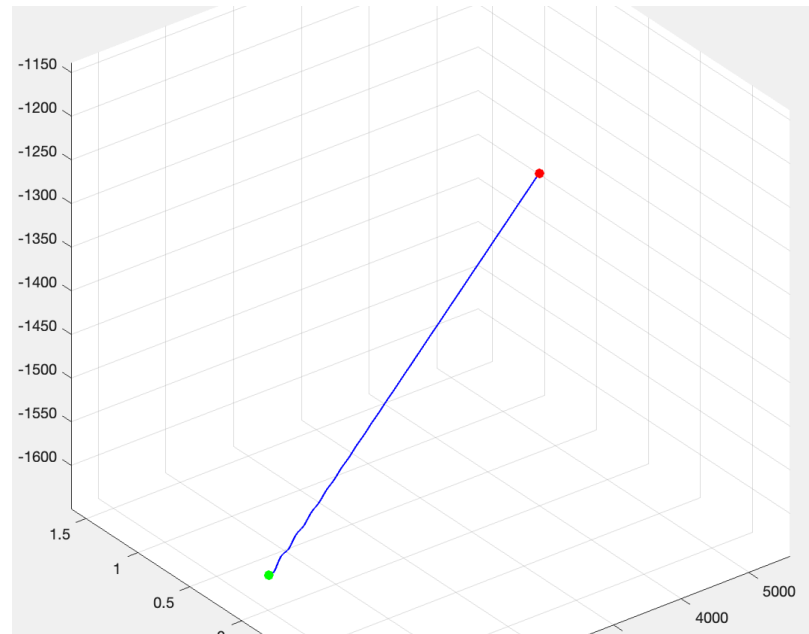


Fig 2.16: 3-D plot for inertial positions $[x,y,z]$ with linear trajectory

Our graph shows a constant y position at 0 and a constant velocity in the positive x direction. We see an almost constant velocity in the z direction as well, with our z position slowly decreasing. There is a quick oscillation in the z-axis that is heavily dampened. The result is a linear trajectory.

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2. Simulation In Initial State 2

When the simulation is run for 200 seconds, our results are:

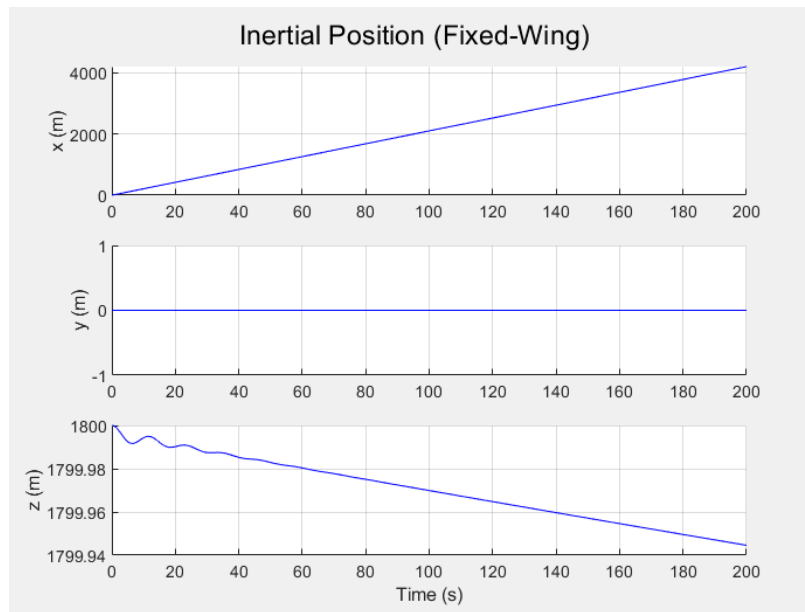


Fig 2.21: Fixed-wing Inertial [x,y,z] Positions

The displacement along the inertial x-axis increased linearly, while changing either not at all, or negligibly, along the inertial y-axis and inertial z-axis. This is consistent with straight and level flight north. There is some minor oscillation in the inertial z-axis due to invoking the phugoid mode, which lasts roughly 60 seconds before being entirely damped away. The reestablished trim is imperfect and so exhibits a very minor descent along the z-axis. There is no lateral disturbance or initial component, so the inertial y-position stays at zero for the duration of the simulation.

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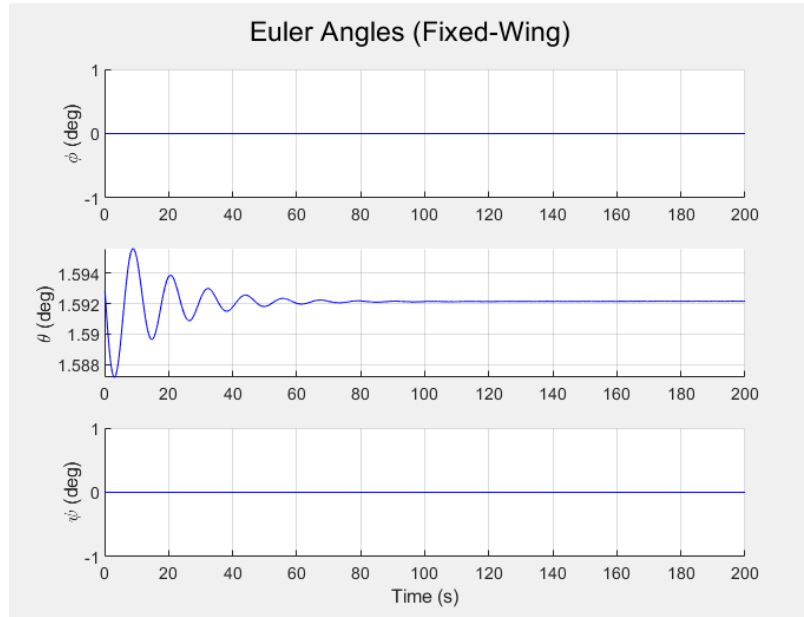


Fig 2.22: Fixed-wing Aircraft Euler angles

There is zero initial component and zero disturbance for both yaw and roll, which is consistent with our given state vector. The pitch oscillates with an initial amplitude of 0.004 degrees and decreases to nearly 0 in roughly 60 seconds. This is consistent with the phugoid mode being invoked and gradually damped away.

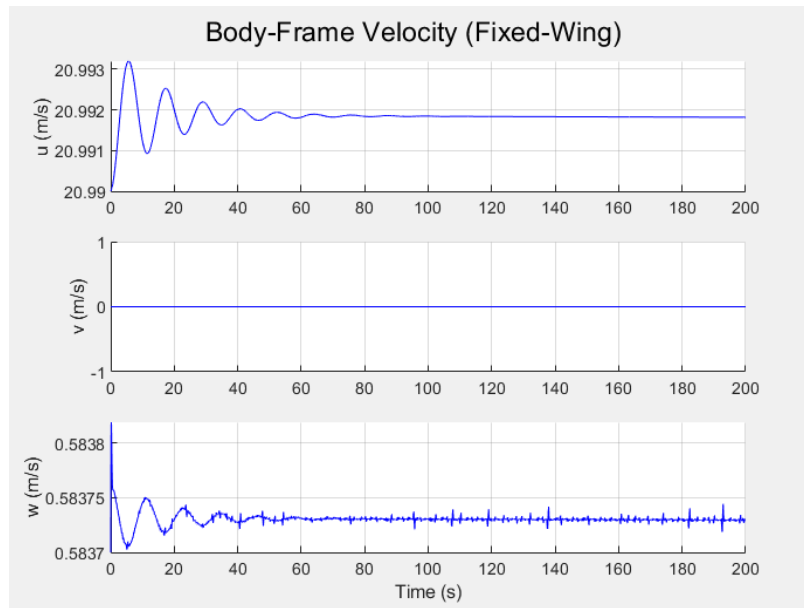


Fig 2.23: Fixed-wing body frame [u,w,v] velocities component

The velocity of the airplane along the direction of its nose experiences oscillations lasting roughly 60 seconds with an amplitude of 0.001 m/s. The velocity in the body y-axis is zero for the entire simulation. The body z-axis experiences minor oscillations of a similar

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magnitude to those along the body x-axis. This is consistent with a plane experiencing a disturbance that invokes the phugoid mode and experiences no lateral disturbance.

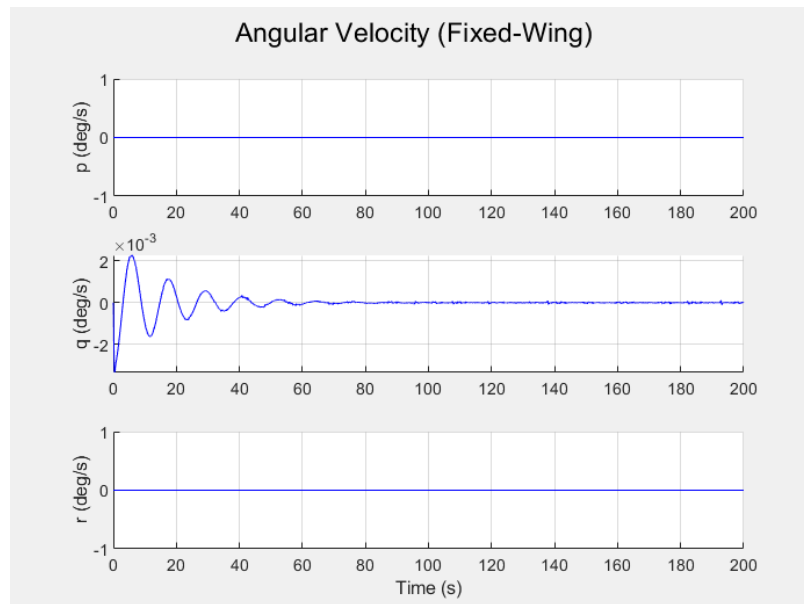


Fig 2.24: Fixed-wing Aircraft Angular Velocities

There is zero angular rate for both p and r , corresponding to the x-axis and y-axis, respectively; there is an initial disturbance in the pitch rate, q , caused by the initial pitch value being nonzero and invoking the phugoid mode, the oscillation of the pitch rate lasts roughly 60 seconds and has an amplitude of roughly 0.002 degrees per second. This is consistent with the initial conditions given and our other data.

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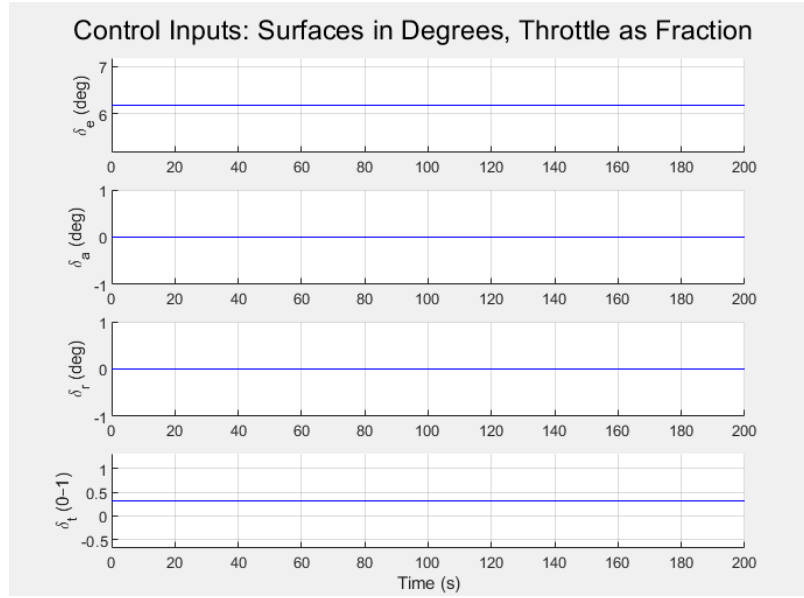


Fig 2.25: Aircraft Control Inputs

The rudder and aileron deflections are zero for the duration of the simulation; the elevator deflection is consistent at just over six degrees for the duration of the simulation, which is consistent with the deflection needed to retain trim. Rudder and aileron deflections are zero because no lateral components are present, which would necessitate their use. The throttle remains at 0.3182 throughout; this only indicates that the pilot has the throttle stick about one-third of the way into its range of motion and doesn't make any adjustment during the simulation. We can assume that the thrust is constant for the duration of the simulation.

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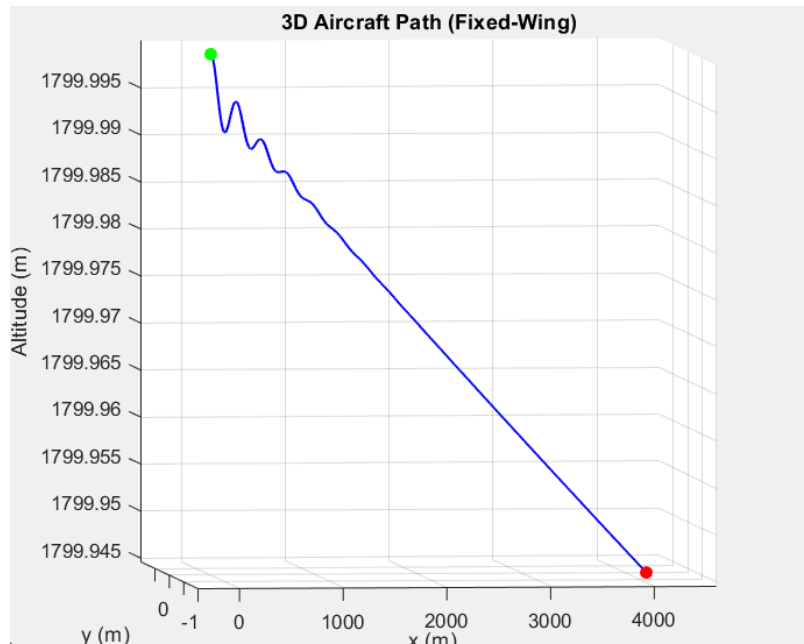


Fig 2.26: 3-D plot for Fixed-wing aircraft with Trajectory

The overall trajectory of the plane is essentially a flat line when observed from the ground; the seemingly downward trajectory is only due to the scaling of the graph in order to make the oscillations visible. The plane starts at an altitude of 1800 meters and ends at roughly 1799.94 meters, which is a negligible difference, over roughly 60 seconds, and having moved roughly 1000 meters along the inertial x-axis, the plane experiences longitudinal oscillations consistent with the invocation of the phugoid mode.

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3. Simulation In Initial State 3

The behavior and trajectory of the aircraft during a simulation spanning 200 seconds are captured below:

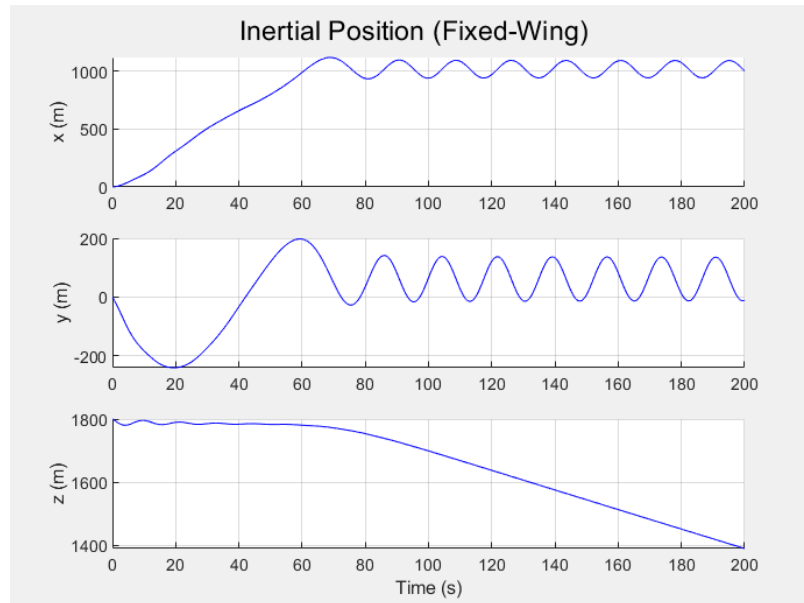


Fig 2.31: Fixed-wing inertial [x,y,z] positions

The displacement along the inertial x-axis increases roughly linearly for about 60 seconds before oscillating with a period of roughly 20 seconds and an amplitude of roughly 50 meters until the end of the simulation. The inertial y-axis displacement exhibits similar behavior, although it decreases to a strongly negative and then strongly positive displacement before entering steady oscillations after 60 seconds. This is consistent with a wide banked turn and subsequent spiral mode, as seen in the trajectory of the aircraft overall in Fig. 2.36 below. The inertial z-axis displacement, altitude, exhibits minor oscillations which are almost entirely gone by 60 seconds, and after 60 seconds, the altitude decreases very linearly at a rate of roughly 2.5 meters per second until the end of the simulation. This is consistent with the downwards trajectory characterizing an unstable spiral mode.

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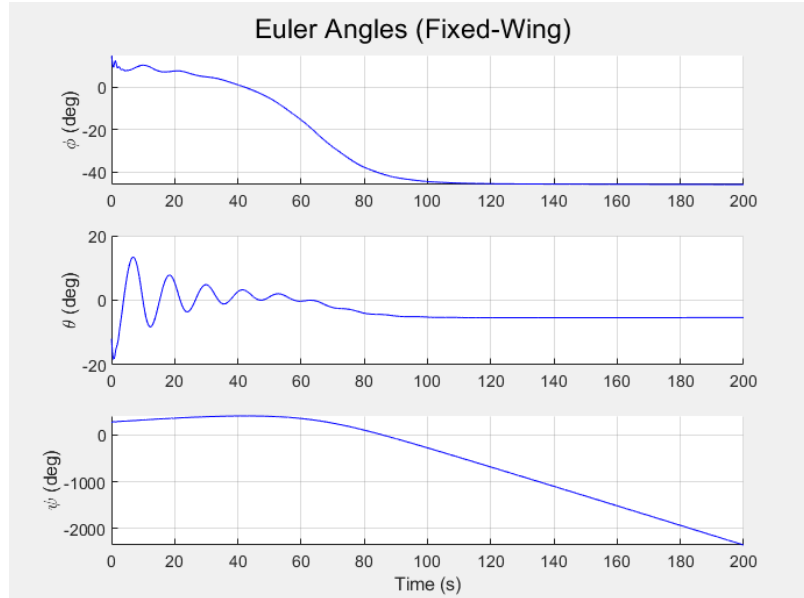


Fig 2.32: Fixed-wing Aircraft Euler angles

All three Euler angles start at or close to zero degrees, consistent with the given initial state vector. Roll and pitch both reach steady state values, whereas yaw's steady state behavior is a linear negative trend at a rate of roughly 25 degrees per second. Because in our coordinate system the angles do not reset when completing a revolution, the angle values can exceed 360 degrees. The -2000 degrees indicates that the aircraft has performed almost 5.5 complete revolutions by the time the simulation ends. This is consistent with the trajectory visible in Fig. 2.36 below. Roll reaches a steady value near negative 42 degrees, and pitch reaches a steady state value of roughly negative five degrees. These are both consistent with a continual banking maneuver, during which the aircraft's wings are not level, and its nose is aimed slightly downward, characteristic of the spiral mode. Notice that although Fig. 2.34 indicates pitch rate settles at a nonzero value, pitch angle reaches a steady value. This is due to the coupling between the pitch and yaw rates in the spiral mode; more explanation regarding this phenomenon is present in that section.

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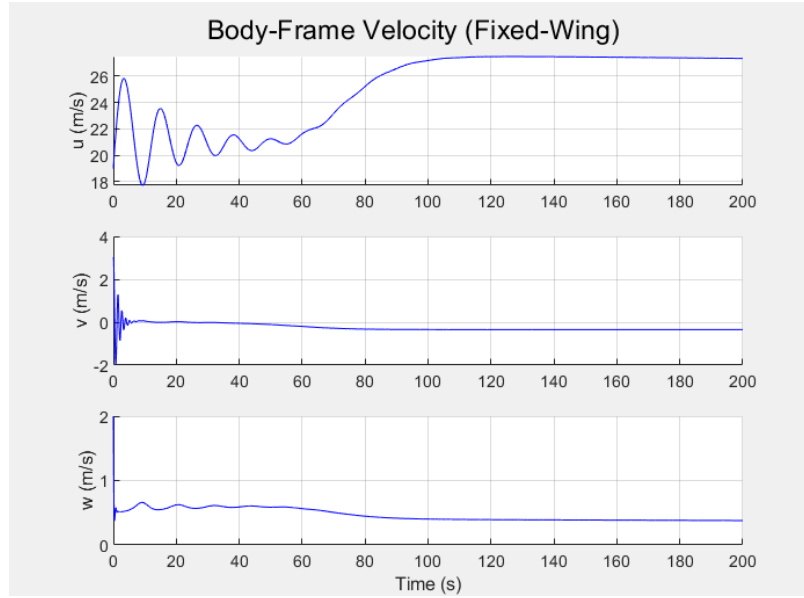


Fig 2.33: Fixed-wing bodyframe $[u,v,w]$ velocities component

The oscillations in $u(t)$ are roughly 10 seconds, whereas the oscillations in $v(t)$ are almost completely gone by seven seconds into the simulation. These correspond to the phugoid and short period modes respectively, $w(t)$ also experiences minor oscillations following a huge initial disturbance. After roughly 60 seconds, the oscillations of $u(t)$ are almost completely gone, and it increases to nearly 28 meters per second and stays there for the rest of the simulation. This is consistent with the steady banked turn and the trade in altitude for velocity that occurs when the aircraft descends from its first loop, as seen in Fig. 2.36 below.

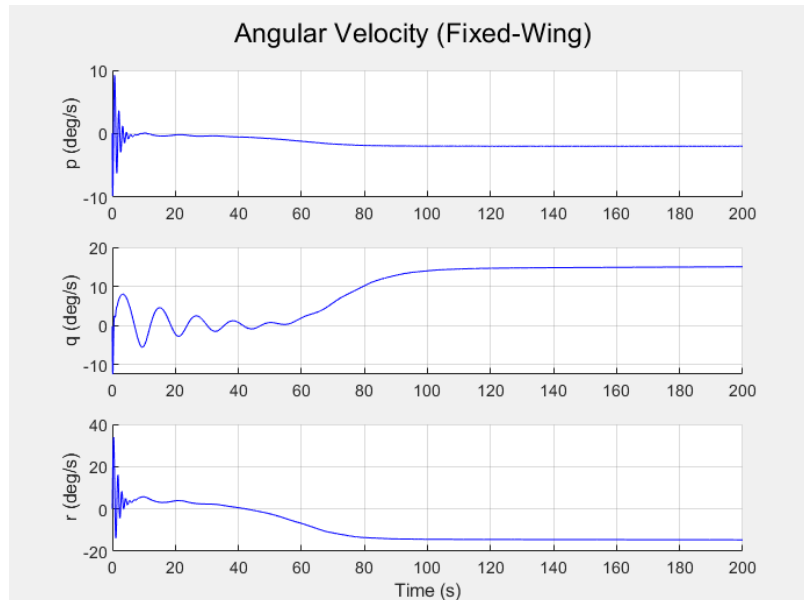


Fig 2.34: Fixed-wing Angular velocities $[p,q,r]$ components

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The roll rate experiences a large initial spike and subsequent rapid decay, consistent with the aircraft initially entering the banking maneuver; after roughly seven seconds, the roll rate has largely stabilized, and by 100 seconds is constant at roughly negative two degrees. The pitch rate shows rapid oscillations with an initial amplitude of roughly eight degrees; these damped oscillations approach zero, but then increase again and by 100 seconds remain steady at 15 degrees. This is not intuitive, as one might expect pitch to return to zero degrees, or continually oscillate, lest the aircraft do loops in the air. However, the nonzero pitch rate is consistent with an airplane in a constantly turning coordinate frame. This nonzero pitch rate is due to coupling between pitch and yaw in the spiral mode. The rate of change of pitch is $q\cos(\phi) - r\sin(\phi)$, hence why the steady state yaw rate is negative, the steady state pitch rate, because both cancel each other, and hence pitch reaches a steady state as seen in Fig 2.32. The yaw rate oscillates for the first seven seconds, but then decreases until reaching a steady -15 degrees, consistent with a continual banked turn characteristic of the spiral mode.

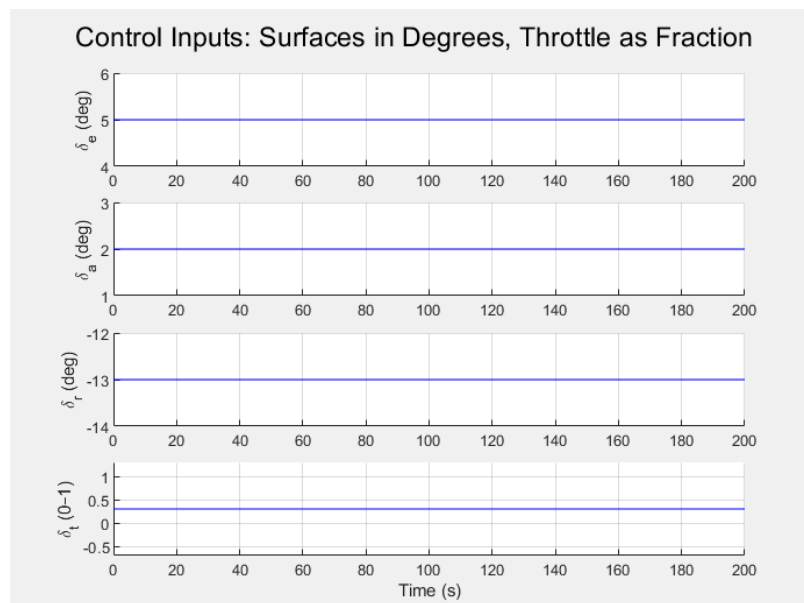


Fig 2.35: Fixed-wing Control Inputs

All surface controls and throttle remain at their initial values for the duration of the simulation. Although there are nonzero initial control inputs, there is still no control law; hence, the control inputs do not change dynamically in response to the aircraft's behavior. If the proper control law were implemented, it is possible that the unstable spiral trajectory could be recovered, but because the initial controls do not counteract it and do not adjust dynamically, it eventually enters a steady downwards spiral.

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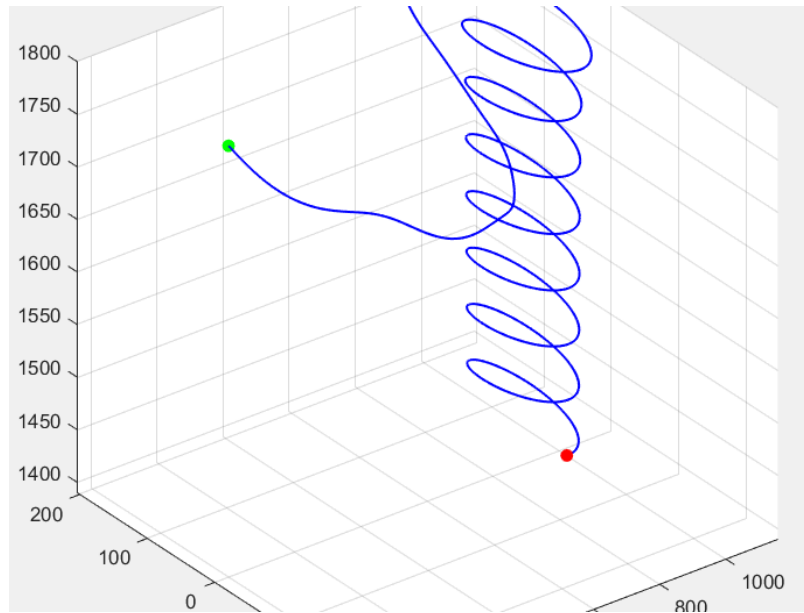


Fig 2.36(a): Fixed-wing 3D aircraft 3D Trajectory Path [Front view]

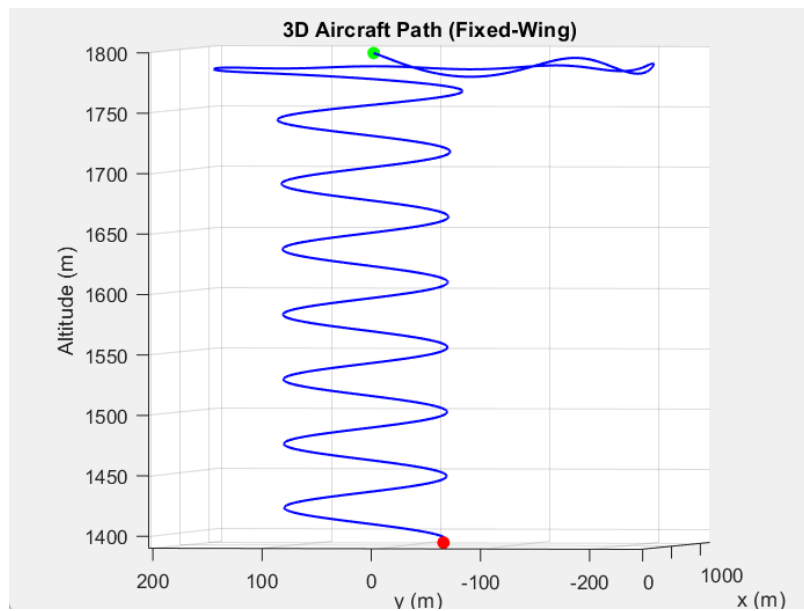


Fig 2.36(b): Fixed-wing 3D aircraft 3D Trajectory Path

The overall trajectory of the aircraft demonstrates an unstable spiral mode. The aircraft enters a banked turn, which it does not roll out of for the duration of the simulation, but persists in steadily. Although the trim condition is stable for the Twistor, the initial conditions are very far from trim, and because Twistor, like many real-life aircraft, has a slightly unstable spiral mode, it

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is not surprising that it exhibits this behavior, and it is very consistent with the initial state vector given as well as all previous figures. This graph also serves to demonstrate the danger of an unstable spiral mode, because the behavior of the aircraft if not corrected will clearly be to spiral all the way into the ground, however, given that the behavior of the craft is very predictable in this mode and that it can last for several minutes, not to mention that it can be recovered with appropriate pilot intervention in most cases, it is understandable why the spiral mode stability of an aircraft is sacrificed to optimize more dangerous modes, such as the dutch roll.

PROBLEM 3: AIRCRAFT EOM WITH DOUBLET

```
% Problem 3
% (Simulating Aircraft Longitudinal Dynamics using ode45) with an elevator doublet input.
clc; clear; close all;
% givens
% Doublet Parameters
doublet_size = deg2rad(15); % Doublet size: 15 degrees, into radians
doublet_time = 0.25; % Duration of each half-pulse
% Define Trim and Initial State (from Problem 2.2/typical linear dynamics)
% x = [u, w, q, theta]' (Perturbation state)
% The trim *perturbation* state x_trim is zero.
x_trim = zeros(4, 1); % Assuming trim state is the origin (zero perturbations)
% From Problem 2.2, the trim elevator input is 0.1079 rad
delta_e_trim = 0.1079;
% Initial state for the simulation (Aircraft starts at trim state)
x0 = x_trim;
% A is the 4x4 stability matrix
A = [-0.1, 0, 0, -9.81;
     0, -0.5, 300, 0;
     0, -0.01, -2, 0;
     0, 0, 1, 0];
% B is the 4x1 control input matrix
B = [0; -0.1; -1.5; 0];
%parameters into a struct to pass to the EOM function
P.doublet_size = doublet_size;
P.doublet_time = doublet_time;
P.delta_e_trim = delta_e_trim;
P.x_trim = x_trim;
P.A = A; % Placeholder matrix
P.B = B; % Placeholder matrix
%tolerance
options = odeset('RelTol', 1e-6, 'AbsTol', 1e-6);
%Short Period (SP) Mode with tspan --> 3 seconds
disp('Starting Short Period Simulation (0 to 3 seconds)...');
t_span_sp = [0 3]; % Short time span to observe the fast Short Period mode
% ode45 call
[t_sp, x_sp] = ode45(@(t, x) AircraftEQMDoulet(t, x, P), t_span_sp, x0, options);
%plots Short Period Results
figure(1);
subplot(2, 2, 1);
plot(t_sp, x_sp(:, 1)); title('Perturbation Velocity u (m/s)'); ylabel('u');
grid on;
subplot(2, 2, 2);
plot(t_sp, x_sp(:, 2)); title('Perturbation Velocity w (m/s)'); ylabel('w');
grid on;
subplot(2, 2, 3);
plot(t_sp, rad2deg(x_sp(:, 3)));
title('Pitch Rate q (deg/s)');
ylabel('q');
xlabel('Time (s)');
grid on;
subplot(2, 2, 4);
plot(t_sp, rad2deg(x_sp(:, 4))); title('Pitch Angle \theta (deg)');
ylabel('\theta');
xlabel('Time (s)');
```

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```

grid on;
title('Aircraft Longitudinal Response to Elevator Doublet (Short Period)');
% Phugoid Mode (phu) with tspan --> 100 seconds
disp('Starting Phugoid Simulation (0 to 100 seconds)...');
t_span_phu = [0 100]; % Long time span to observe the slow Phugoid mode
% ode45 call
[t_phu, x_phu] = ode45(@(t, x) AircraftEQMDoulet(t, x, P), t_span_phu, x0, options);
% plots Phugoid Results
figure(2);
subplot(2, 2, 1);
plot(t_phu, x_phu(:, 1)); title('Perturbation Velocity u (m/s)'); ylabel('u'); grid on;
subplot(2, 2, 2);
plot(t_phu, x_phu(:, 2)); title('Perturbation Velocity w (m/s)'); ylabel('w'); grid on;
subplot(2, 2, 3);
plot(t_phu, rad2deg(x_phu(:, 3))); title('Pitch Rate q (deg/s)'); ylabel('q'); xlabel('Time (s)'); grid on;
subplot(2, 2, 4);
plot(t_phu, rad2deg(x_phu(:, 4))); title('Pitch Angle \theta (deg)'); ylabel('\theta'); xlabel('Time (s)');
grid on;
title('Aircraft Longitudinal Response to Elevator Doublet (Phugoid)');
% Aircraft EOM (For Problem 3)
function xdot = AircraftEQMDoulet(t, x, P)
% Calculates the linearized longitudinal EOM for the aircraft with a doublet
% applied to the elevator.
% State vector x = [u, w, q, theta]' (perturbations)
% Control input is delta_e (perturbation from trim)
% Unpack parameters
delta_e_trim = P.delta_e_trim;
doublet_size = P.doublet_size;
doublet_time = P.doublet_time;
A = P.A;
B = P.B;
% 1. Calculate the elevator command delta_e(t) (total value) using the doublet logic
% Equation 7 on lab assignment
if t <= doublet_time
    delta_e_total = delta_e_trim + doublet_size;
elseif t <= 2 * doublet_time
    delta_e_total = delta_e_trim - doublet_size;
else
    delta_e_total = delta_e_trim;
end
% 2. Calculate the perturbation control input, delta_e_pert.
% The linearized EOM is  $\dot{x} = Ax + B\delta_{e,pert}$ , where  $\delta_{e,pert} = \delta_{e,total} - \delta_{e,trim}$ .
delta_e_pert = delta_e_total - delta_e_trim;
% 3. Calculate the state derivative (linearized EOM)
xdot = A * x + B * delta_e_pert;
end

```

1. 3 Second Run Case:

The behavior and trajectory of the aircraft during a simulation spanning three seconds are captured below:

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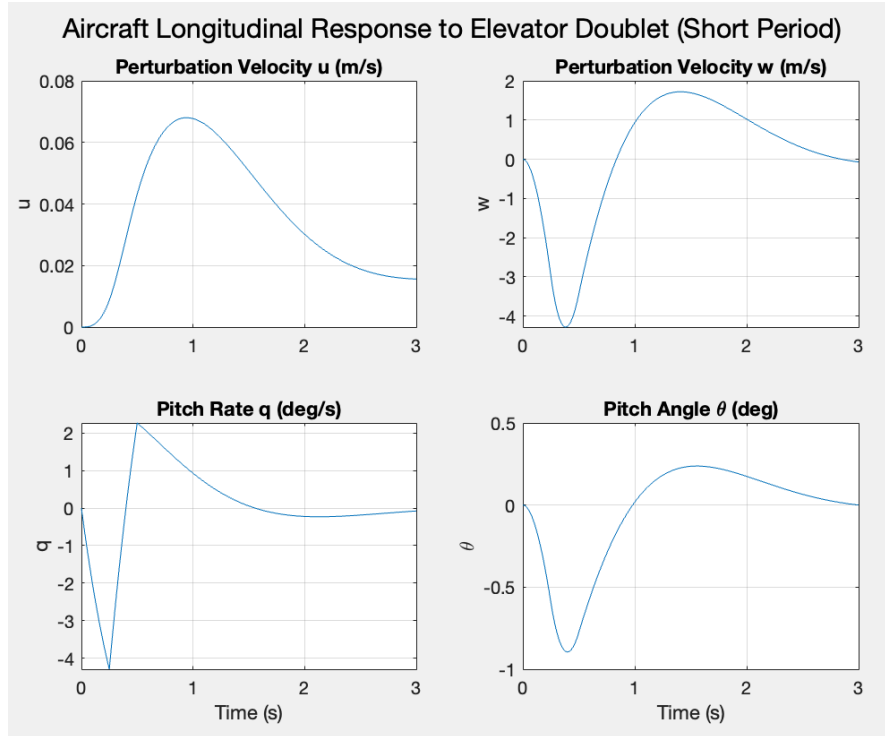


Fig 3.1: Short Period Plots Response showing the perturbation velocity $[u, w]$ and pitch rate(q) and Pitch angle (θ)

Our graphs of the first three seconds of flight are for velocity $[u, w]$ and pitch rate (q) and pitch angle (θ). As perturbations cause the velocity to rise exponentially to maximum until $t = 1$ sec approx. and decays from 1 to 3 seconds. The perturbation velocity w increases in magnitude from 0 to -4 m/s in about 0.5 sec time frame in negative direction while it starts to rise after it 0.5s and goes to $+1.5$ m/s and then decays to become constant at $t = 3$ sec. The pitch rate shows the spikes where it changes direction and falls from 0 to -4 deg/s and then rises to 2 deg/s within 1s time frame that reflects the short period mode as how quickly it comes back to its normal position that happens when it approaches 3s. Pitch angle shows the similar patterns as it changes the w component of velocity is negative that shows the pitch angle is negative (nose down) and as it approaches positive (nose up) and the aircraft starts to climb up.

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2. 100 Second Run Case:

When the simulation is run for 100 seconds, our results are:

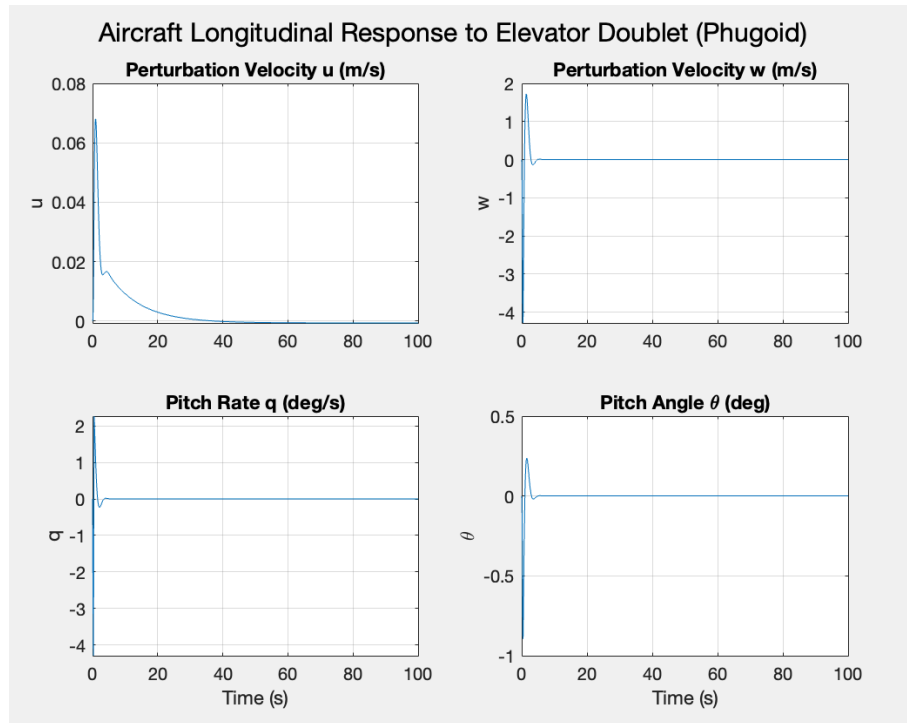


Fig 3.2: Phugoid Plots Response showing perturbation velocity $[u, w]$ and pitch rate(q) and pitch angle(θ)

Phugoid as it takes longer to dampen in and reach a steady state lasts for a longer time. Looking at the time span $[0\ 100]$ that shows the u velocity component shows the sharp rise as aircraft experience perturbations and that allows the aircraft to take quite longer than compared to short period to reach stable motion and where the u component of velocity approaches 0 m/s . The perturbation in the w velocity component shows that pretty quickly approx. $t = 5\text{s}$ returns to 0m/s that shows it affects the u component of velocity for a longer time period compared to w . Pitch rate reflects the similar response as the perturbation w velocity component and that shows the aircraft rise and fall in vertical direction diminishes in about 5s and that does not contribute to phugoid mode any longer while the dominant component is the u component of velocity. Lastly, the pitch angle changes reflect the nose up and down position of the aircraft while in motion, when perturbation u and w velocity components act on it. The small magnitude perturbation in the u velocity component did not tend to affect the aircraft in phugoid mode for longer. As other components return to steady state pretty quickly showing diminishing effect of phugoid mode and just the forward airspeed of aircraft solely acting on it.

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Name	Plan	Model	Experiment	Results	Report	Code	ACK
Nico Galindo	1	1	1	1	1	1	X
Michael Mcallister	1	1	1	1	1	1	X
Alexander Lu	1	1	1	1	1	1	X
Ritik Sarraf	1	1	1	1	1	1	X

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